Image Dithering: Notes

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1 What is Dithering?

1.1 Quantisation

A digital image stores each pixel as a value from a limited set of intensities. In the real world, light is continuous, and the human eye can perceive millions of colours. Whereas a display (or file format) can only represent a *reduced* palette of colors, every pixel must be mapped to one of the available values. This is known as *quantisation*.

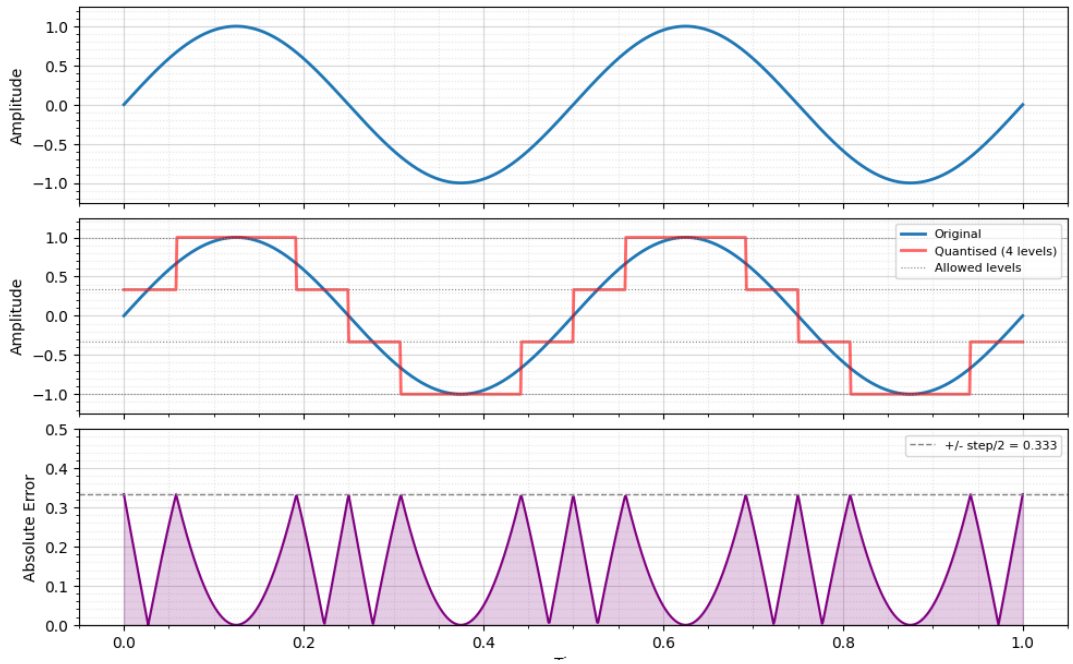


Figure 1: Quantisation of a continuous wave. The original signal (blue) quantised to 2 bits = 4 levels (red).

If we look at images, let a single channel of an image be a function $I(x, y) \in [0, 255]$. A quantiser with L levels snaps each value to the nearest of L evenly spaced values:

$$\Delta = \frac{255}{L-1}, \qquad q(v) = \Delta \cdot \text{round}\Big(\frac{v}{\Delta}\Big). \tag{1}$$

The difference between an input value and its quantized value is called the *quantisation error*, denoted by $e = v - q(v)$, and always satisfies $|e| \leq \Delta/2$. For a colour image the quantiser is applied to the red, green and blue channels independently, giving L^3 possible colours (Table 1).

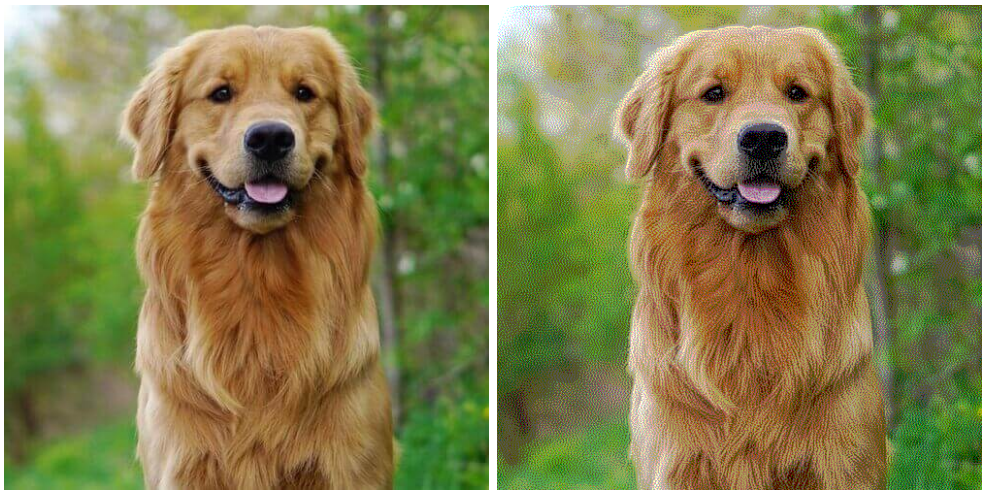
Table 1: Palette size as a function of levels per channel L.

L	Grayscale shades	Colours (L^3)	Step Δ
2	2	8	255
3	3	27	127.5
4	4	64	85
6	6	216	51
256	256	16,777,216	1

As L increase, the quantisation error decreases hence the amount of information lost decreases. However, the number of colours grows cubically with L, so any small increase in L can drastically increase a file size. This is why many image formats use a small palette of colours, and why dithering is important.

1.2 Dithering

Dithering is the family of techniques that arranges the available values spatially so that, viewed from a distance, the eye averages neighbouring pixels and perceives intermediate tones that the palette cannot represent directly. In short, dithering trades spatial resolution for tonal resolution. Instead of mapping each pixel independently to the nearest palette colour, dithering *remembers* the quantisation error and distributes it to neighbouring pixels, such that the average tone is preserved.



(a) Original (b) Dithered

Figure 2: Original versus dithered

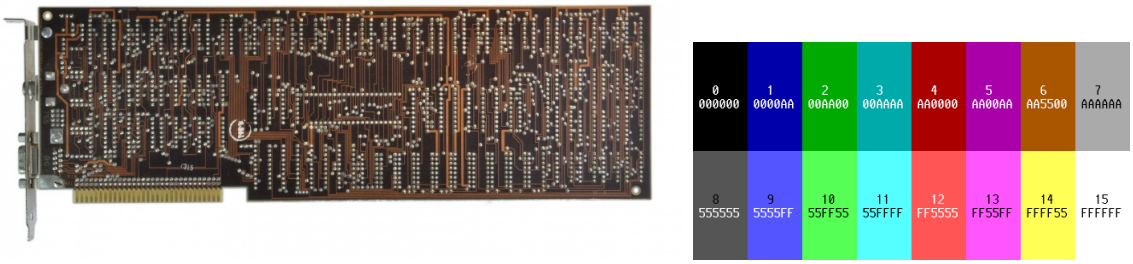
Dither is routinely used in processing of both digital audio and visual data, and is a key technique in computer graphics. The result is almost always a more faithful reproduction of the original image. We'll focus on the visual case, where dithering is used to reduce the number of colours in an image.



Figure 3: Dithering of the Mona Lisa. The original image (left) is quantised to 2 levels per channel and then dithered (right).

2 Historical Context

Early display hardware could show only a handful of colours at once. The IBM Colour Graphics Adapter (CGA), introduced in 1981 with the IBM PC, is the classic example. This was the first monitors in the market to display color. Image resolution was low, and the palette was limited to 16 colours. Its 16-colour palette is built from four on/off signals: red, green, blue and an intensity bit (RGBI). Sixteen colours need only four bits per pixel, since $2^4 = 16$.



(a) IBM CGA card

(b) 16 colour palette on a CGA monitor

Figure 4: The IBM Colour Graphics Adapter (CGA) and its 16-colour palette.

This should not be confused with "16-bit colour" (High Color) used in modern systems, which uses 16 bits per pixel and yields $2^{16} = 65{,}536$ colours (Table 2).

Table 2: Bits per pixel versus number of distinct colours.

Name	Bits per pixel	Colours
CGA palette	4	16
"16-bit" High Color	16	65,536
24-bit True Color	24	16,777,216

CGA offered several video modes, each with a different resolution and palette size suited to different applications with different memory constraints:

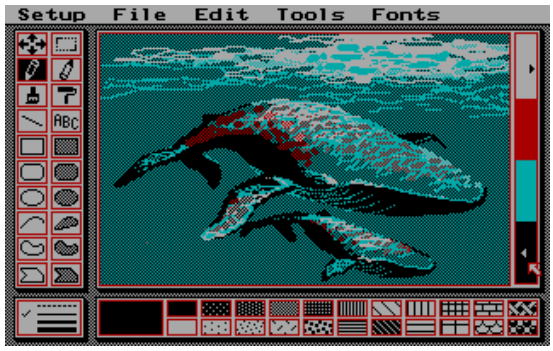
- a 160x100 mode with 16 colours
- a 320x200 mode with 4 colours
- a 640x200 mode with 2 colours

Each mode was a compromise between resolution and colour depth. For some programs, they were small enough to make use of the 16-colour mode. But the limited palette for programs with a larger resolution was a problem. Programmers therefore mixed pixel patterns to imitate the missing colours, which is a major reason dithering became widespread. In the 1980s, dithering was used in many games and graphics programs to simulate more colours than the hardware could display.

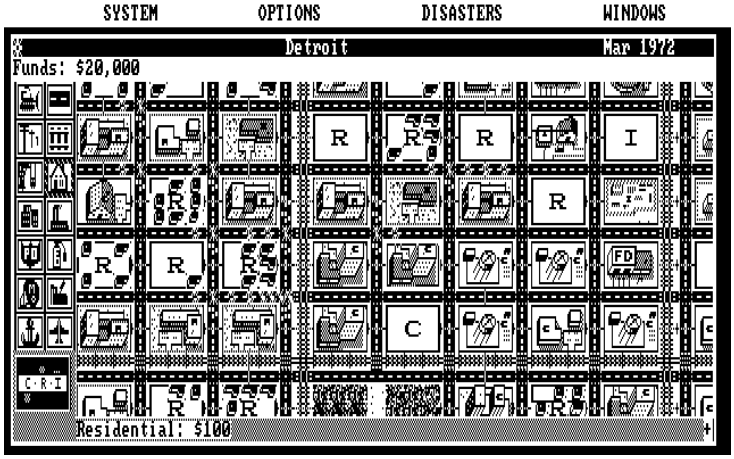
Take a look at Figure 5. The three images are from the same era, but they use different CGA modes. For example, the game 5a, the original Pac-Man, uses the 160x100 16 colour mode. The game makes use of the full palette, each ghost is a different colour, each fruit is a different colour, and the yellow Pac-Man really pops out of the screen. However, although in this mode the resolution is limited, the user is still able to experience the game just fine and play it without any problems. While the program 5b uses the 320x200 4 colour mode, PCPaint is designed to create images, and the user is able to select the colours they want to use. At that the time, the 4 colour mode was a compromise between resolution and colour depth. Finally, 5c uses the 640x200 monochrome mode. It's a city-building simulation game where the user is able to build their own city. It's a very complex game, to display all the information on the screen, the program uses the 640x200 monochrome mode. However, beause of the limited palette, the program uses dithering to simulate gray tones and a non-square pixel ratio that deforms the fonts. The use of dithering helps to make up for the lack of colours and makes the game more visually appealing.



(a) PakuPaku in 160x100 16 color mode



(b) PCPaint in 320×200 3rd palette low intensity



(c) SimCity in 640×200 monochrome

Figure 5: The IBM Colour Graphics Adapter (CGA) and its 16-colour palette.

3 Use Cases: Early Game Graphics

Console and arcade artists faced the same constraint: a small palette and low resolution. But are tasked with creating a visually rich experience. The Sega Mega Drive (Genesis) offered a 9-bit palette of 512 colours, of which only a limited number could be shown at once. Artists in games such as *Sonic the Hedgehog* used checkerboard patterns of two colours; on a CRT television connected by composite video, the signal blurred the pattern into an apparent third colour or a semi-transparent effect (a commonly cited example is the waterfalls of Green Hill Zone).

Verify details and cite a source.



Figure 6: Sonic the Hedgehog (1991) on the Sega Mega Drive

Dithering also served monochrome displays: Atkinson's algorithm (Section 5.3) was written for the original Apple Macintosh, whose screen was strictly one bit per pixel.

4 The Primitive Approach: Nearest Colour

You might think why do we need dithering? Why can't we just quantise the image to the nearest colour in the palette? The simplest strategy maps every pixel independently to the closest palette colour. With palette P and pixel $p = (r, g, b)$,

$$\text{map}(p) = \arg\min_{c \in P} \sqrt{(r - c_r)^2 + (g - c_g)^2 + (b - c_b)^2}. \tag{2}$$

For two grayscale levels this reduces to thresholding at 128.

Take a look at the example in Figure 7. The original image of an apple is quantised to different levels of quantisation, 10, 5, and 2. You can see that as the number of quantisation levels decreases, the image loses more and more information. At level 10, the image is still recognisable, but at level 5, the image is starting to lose its details. Notice the banding at the center of the apple as well as on the leaf, the difference shades of green or red are lost, its more apparent where certain pixels are mapped to the same colour and its easier to recognise the grainy texture of pixalized images. At level 2, its evident that the image has lost a lot of information, its a day and night difference compared to the original image. Because of the quantisation error, the original color of the apple is lost, its misrepresented and is an entirely different shade of red, and the image looks flat and unnatural.

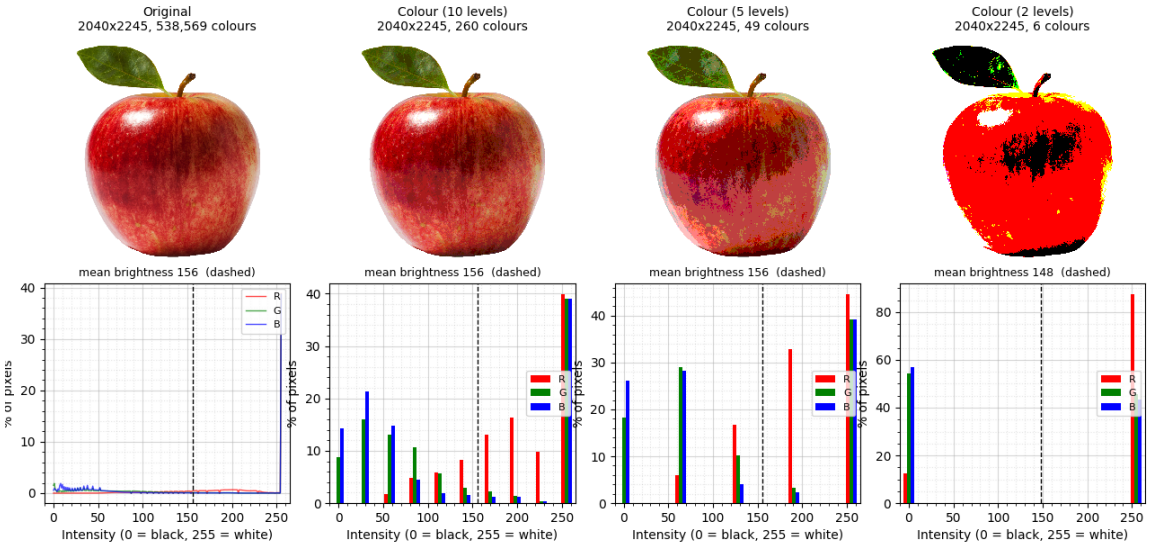


Figure 7: Nearest colour mapping of an apple to different levels of quantisation

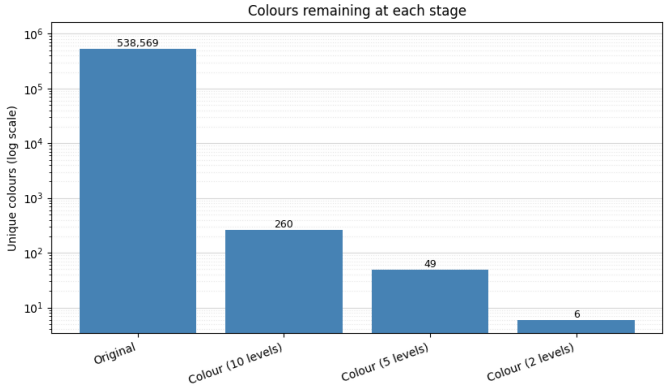
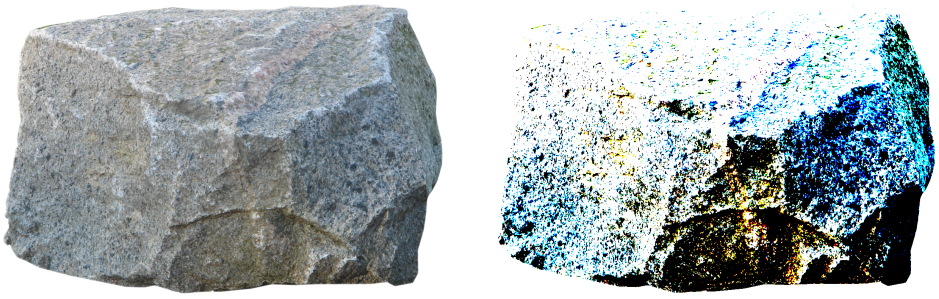


Figure 8: Bar chart of the number of distinct colours in the original and the quantised images

The bar chart in Figure 8 shows the number of distinct colours in the quantised images drops logarithmically as the number of quantisation levels decreases. It terms of numbers, the original image has 538,569 distinct colours, while the all quantised images have less literally than 0.048% of the original number of distinct colours. Although its apparent for levels 5 and 2, the color information is significantly reduced, its actually not as apparent in level 10, the number of distinct colours is still reduced by 99.95%, but the image still looks recognisable and retains a lot of its original information. This ultimately is an optical illusion and has to do with how the human eye perceives colour and light, the image looks recognisable but we know the color information is significantly reduced, what we perceive as a single shade of colour is actually a combination of different distinct colours.

Here's another example of nearest colour mapping in Figure 9. The original image of a rock is mostly grey, with some brown and blue tones. When mapped to a 16-colour palette, the image loses a lot of its original information; grey areas are mapped to either black or white, and those small specks of brown and blue tones are now very apparent.



(a) Original image of a rock (b) Mapped to a 16-colour palette

Figure 9: Nearest colour mapping of a rock

Nearest-colour mapping discards the error entirely. Smooth gradients collapse into flat regions with hard edges (*banding* or *contouring*), and large areas of similar tone become a single solid colour. The algorithms in Section 5 fix this by *remembering* the error.

Insert a threshold-versus-dithered comparison figure.

5 Dithering Algorithms

There are many ways to dither an image. The simplest is to add random noise to each pixel before quantisation, which breaks up the banding but produces a grainy texture. A more sophisticated approach is to use a *dither matrix* to distribute the quantisation error in a regular pattern. The most common method is *error diffusion*, which spreads the error to neighbouring pixels that have not yet been processed.



(a) Original



(b) Floyd-Steinberg



(c) Atkinson



(d) Jarvis–Judice–Ninke



(e) Sierra

Figure 10: Different Dithering Algorithms applied to the same image.

5.1 Error diffusion

All four algorithms below are *error-diffusion* methods. Pixels are processed in scan order. For the current pixel at (x, y):

$$\text{old} = I(x, y), \qquad \text{new} = q(\text{old}), \qquad e = \text{old} - \text{new}, \tag{3}$$

$$I(x + d_x,\, y + d_y) \mathrel{+}= \frac{w}{D}\, e \quad \text{for each } (d_x, d_y, w) \text{ in the kernel.} \tag{4}$$

The kernel lists which not-yet-processed neighbours receive a share of the error, with integer weights w and a divisor D. In the matrices below, $*$ marks the current pixel and $\cdot$ marks pixels that are never modified.

Our implementation uses *serpentine* scanning: the scan direction alternates every row (and the kernel is mirrored accordingly), which avoids the diagonal "worm" artefacts of always scanning left to right.

5.2 Floyd–Steinberg

Published by Floyd and Steinberg in 1976, this is the most widely used algorithm. It spreads the full error over four neighbours:

$$\frac{1}{16}\begin{pmatrix} \cdot & * & 7 \\ 3 & 5 & 1 \end{pmatrix} \tag{5}$$

5.3 Atkinson

Developed by Bill Atkinson at Apple for the original Macintosh, this kernel spreads the error over six neighbours but divides by 8, so only $6/8 = 75\%$ of the error is propagated:

$$\frac{1}{8}\begin{pmatrix} \cdot & * & 1 & 1 \\ 1 & 1 & 1 & \cdot \\ \cdot & 1 & \cdot & \cdot \end{pmatrix} \tag{6}$$

Discarding a quarter of the error gives higher contrast: highlights and shadows tend to saturate to pure white and black, while mid-tones keep crisp, less noisy detail.

5.4 Jarvis–Judice–Ninke

This kernel spreads the error over twelve neighbours across three rows:

$$\frac{1}{48}\begin{pmatrix} \cdot & \cdot & * & 7 & 5 \\ 3 & 5 & 7 & 5 & 3 \\ 1 & 3 & 5 & 3 & 1 \end{pmatrix} \tag{7}$$

The wider spread produces smoother results with finer texture, at the cost of more computation per pixel.

5.5 Sierra

The three-row Sierra kernel is a lighter relative of Jarvis–Judice–Ninke, using ten neighbours and a power-of-two divisor:

$$\frac{1}{32}\begin{pmatrix} \cdot & \cdot & * & 5 & 3 \\ 2 & 4 & 5 & 4 & 2 \\ \cdot & 2 & 3 & 2 & \cdot \end{pmatrix} \tag{8}$$

5.6 Comparison

Table 3: Summary of the error-diffusion kernels used.

Algorithm	Neighbours	Divisor D	Error propagated
Floyd–Steinberg	4	16	100%
Atkinson	6	8	75%
Jarvis–Judice–Ninke	12	48	100%
Sierra	10	32	100%

6 Conclusion

A Dithered Image Gallery

Verify publication years and add references for each algorithm.

Add side-by-side result figures for the same source image.

Summarise your findings. Observations from the analysis tool that may be worth confirming on real images: (1) a dithered image's histogram collapses onto L values

Table 4: Summary of all example images used.

Source	Size	Colours	Gray lvls	Mean	Std	Min	Max
Apple.png	2040x2245	538,569	249	156.37	85.12	7	255
Banana.jpg	3300x3300	58,356	255	246.24	19.66	1	255
Blackwhite-Wallpaper.jpg	1920x1080	256	256	113.16	111.32	0	255
Bruno-Mars.jpg	916x639	124,696	255	73.26	72.85	1	255
Canelo.jpg	451x680	36,513	244	73.80	55.71	0	246
Cat.jpg	554x554	90,238	246	118.83	57.10	0	251
Chun-Li.png	820x523	73,098	256	191.30	89.01	0	255
Eagle.jpg	1536x1155	138,380	256	136.80	60.89	0	255
Emoji.png	1024x1024	38,921	215	197.30	54.31	41	255
Flowers.jpg	1920x1200	207,874	243	109.21	61.14	0	243
GitHub.jpg	474x484	506	256	212.56	79.19	0	255
Golden-Retriever.jpeg	500x500	94,919	256	118.76	37.86	0	255
Greek-Statue.png	1024x1024	53,965	255	63.58	59.70	0	254
Iron-Man.jpg	840x930	46,620	256	220.40	63.64	0	255
KFC-Logo.png	1219x914	14,477	236	152.23	92.63	20	255
Lanterns.jpg	2160x1080	76,346	256	101.17	57.59	0	255
Mario.jpg	1350x1080	206,464	256	195.13	79.48	0	255
Mona-Lisa.jpg	1057x1600	272,610	256	60.15	64.78	0	255
Parade.jpg	1500x1000	527,062	256	139.82	63.42	0	255
Pen.jpg	1000x1000	30,819	256	240.88	46.05	0	255
Pokémon-Legends-ZA.jpg	1424x800	158,072	256	75.48	45.46	0	255
Rock.png	1280x960	166,146	231	181.19	63.13	25	255
Samus.png	1529x1034	159,799	256	214.27	74.93	0	255
Sonic.png	1209x1677	16	15	183.07	92.06	0	255
Sphere.jpg	630x630	23,319	210	143.77	70.64	44	255
Starry-Night.jpg	565x353	133,414	256	110.87	66.03	0	255
Tennis-Balls.jpg	1600x900	71,200	235	182.29	36.45	1	236
WiFi.jpg	474x484	10,549	213	194.27	80.25	42	255
Windows-XP-wallpaper-4K.jpg	2560x1707	125,061	219	128.74	42.59	21	239
Yahoo.png	920x920	26,412	213	107.70	79.71	43	255